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American Petroleum Institute (API) Categories CK-4 and FA-4 Diesel Engine Oil Introduction

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### Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

### Contents

This document announces the release of two new American Petroleum Institute (API) categories for diesel engine oils. These oils have been designed for use in on-highway and off-highway applications using ultra-low sulfur diesel (ULSD) fuel.

- API category CK-4 oils were released in December 2016 and include all viscosity grades of CJ-4 oils. As higher engine temperatures and lower emissions are a continuing trend, improved oils were required. CK-4 has improved oxidation resistance, shear stability, and aeration control. API category CK-4 is backward compatible with API category CJ-4 oil technology.
- API category FA-4 oils were also released in December 2016, primarily as a 10W-30 viscosity grade. FA-4 oils carry the same improved oxidation resistance, shear stability, and aeration control of CK-4 oils. FA-4 oils also reduce greenhouse gas emissions (increased fuel economy, dependent on duty cycle) by lowering high temperature/high shear (HTHS) viscosity. Lower HTHS results in directionally thinner oils films lubricating metal-metal interfaces. API category FA-4 oils are **not** backwards compatible and are **only** to be used when directed by the Lubricating Oil Recommendations and Specifications procedure in Section V of corresponding Owners or Operation and Maintenance Manual to use CES 20087 engine oils.

#### **⚠ CAUTION ⚠**

**API category FA-4 oils is never to be used in unapproved engines as engine damage will result.**

#### **Cummins Engineering Standard (CES)**

The API oil category, or any international oil certification, are **not** able to be used to select the appropriate engine oil for a Cummins® engine. **Only** use oils that meet the appropriate Cummins® Engineering Standard (CES). Table 1 below provides a comparison between CES and other oil regulatory classifications for reference **only**. A list of oils that meet the appropriate CES for the engine can be obtained from a local Cummins® distributor or can be found on QuickServe® Online. Reference weblink below.

**Note :**  
[https://QuickServe.Cummins.com/qs3/qsol/service/serviceproducts/oil\\_registration.html](https://QuickServe.Cummins.com/qs3/qsol/service/serviceproducts/oil_registration.html)  
([https://QuickServe.Cummins.com/qs3/qsol/service/serviceproducts/oil\\_registration.html](https://QuickServe.Cummins.com/qs3/qsol/service/serviceproducts/oil_registration.html))

**Note :** CES 20086 is recommended for use anywhere CES 20081 was previously recommended. CES 20086 oil can be used in any diesel engines using ULSD fuel. If performing routine oil analysis, a change can possibly be observed in the oil chemistry.

| Table 1, CES Comparison to Other Oil Regulatory Classifications |                      |                                      |
|-----------------------------------------------------------------|----------------------|--------------------------------------|
| CES                                                             | Closest API Category | Closest International Classification |
| CES 20081                                                       | CJ-4                 | ACEA E9 or JAMA DH-2                 |
| CES 20086                                                       | CK-4                 | None                                 |
| CES 20087                                                       | FA-4                 | None                                 |

**Associated Publications**

| Table 2, Recommendations and Specifications |          |                                                    |                                                                                |         |
|---------------------------------------------|----------|----------------------------------------------------|--------------------------------------------------------------------------------|---------|
| Manual Type                                 | Bulletin | Procedure Title                                    | Procedure                                                                      | Section |
| Fluids for Cummins Products Service Manual  | 5411406  | Engine Oil Classifications                         | Refer to Procedure 378-003 (/qs3/pubsys2/xml/en/procedures/00/00-378-003.html) | 4       |
|                                             |          | Diesel Engine Oil Requirements                     | Refer to Procedure 378-005 (/qs3/pubsys2/xml/en/procedures/00/00-378-005.html) | 4       |
| Operation and Maintenance Manual            | Various  | Lubricating Oil Recommendations and Specifications | 018-003                                                                        | V       |

**Document History**

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